

Scotogenic/shared-scalar group factors: UV

Active theory: SM + S + F . The formal T3 legs are identified as $S_1 = i\sigma_2 S^*$ and $S_2 = S$, so the physical scalar is counted once in loops.

$$j = \frac{d-1}{2}, \quad C_2(d) = \frac{d^2-1}{4}, \quad T(d) = \frac{d(d^2-1)}{12}.$$

In every comparison table, $\mathcal{S}$ denotes the RGE structure multiplying the displayed group factor in $16\pi^2\beta$.

Notation key

symbol	interaction / definition
S	physical inert scalar doublet, $S \sim (2, +1/2)$
$\tilde{S}$	$i\sigma_2 S^*$; formal matching leg $S_1 = \tilde{S}$, while $S_2 = S$
F	neutral singlet or triplet Majorana fermion
h	$LFS + \text{h.c.}$
M_F	heavy-fermion Majorana mass
T_ν	$\text{Tr}(h^\dagger h)$
m_S^2	$m_S^2 S^\dagger S$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
λ_2	$\frac{1}{2}\lambda_2 (S^\dagger S)^2$
λ_3	$\lambda_3 (H^\dagger H)(S^\dagger S)$
λ_4	$\lambda_4 (H^\dagger S)(S^\dagger H)$
λ_5	$\frac{1}{2}\lambda_5 [(H^\dagger S)^2 + \text{h.c.}]$

One-loop beta functions in the physical one-scalar theory

$\beta_{g_1}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
g_1^3	7

$\beta_{g_2}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
g_2^3	–3

$\beta_{g_3}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
g_3^3	–7

$\beta_{Y_u}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$(Y_d(Y_d)^\dagger Y_u)_{ij}$	$-\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_u)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_u)_{ij}$	$-\frac{9}{4}$

$g_3^2(Y_u)_{ij}$	-8
$g_1^2(Y_u)_{ij}$	$-\frac{17}{12}$
$T(Y_u)_{ij}$	1

$\beta_{Y_d}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$(Y_d(Y_d)^\dagger Y_d)_{ij}$	$\frac{3}{2}$
$(Y_u(Y_u)^\dagger Y_d)_{ij}$	$-\frac{3}{2}$
$g_2^2(Y_d)_{ij}$	$-\frac{9}{4}$
$g_3^2(Y_d)_{ij}$	-8
$g_1^2(Y_d)_{ij}$	$-\frac{5}{12}$
$T(Y_d)_{ij}$	1

$\beta_{Y_e}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$(h(h)^\dagger Y_e)_{ij}$	$\frac{1}{2}$
$(Y_e(Y_e)^\dagger Y_e)_{ij}$	$\frac{3}{2}$
$g_2^2(Y_e)_{ij}$	$-\frac{9}{4}$
$g_1^2(Y_e)_{ij}$	$-\frac{15}{4}$
$T(Y_e)_{ij}$	1

$\beta_h^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$T_\nu(h)_{ij}$	1
$(Y_e(Y_e)^\dagger h)_{ij}$	$\frac{1}{2}$
$(h(h)^\dagger h)_{ij}$	$\frac{3}{2}$
$g_2^2(h)_{ij}$	$-\frac{9}{4}$
$g_1^2(h)_{ij}$	$-\frac{3}{4}$

$\beta_{M_F}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$((h)^T(h)^* M_F)_{ij}$	1
$((M_F)^T(h)^\dagger h)_{ij}$	1

$\beta_{m_S^2}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$m_S^2(2T_\nu + 6\lambda_2)$	1
$g_2^2 m_S^2$	$-\frac{9}{2}$
$g_1^2 m_S^2$	$-\frac{3}{2}$
T_ν	-16

$\beta_{\lambda_1}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$\lambda_1(12 \text{Tr}(Y_d(Y_d)^\dagger) + 4 \text{Tr}(Y_e(Y_e)^\dagger) + 12 \text{Tr}(Y_u(Y_u)^\dagger))$	1
g_2^4	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_1)$	1
$g_2^2\lambda_1$	-9
g_1^4	$\frac{3}{4}$
λ_3^2	4
$\lambda_3\lambda_4$	4
λ_4^2	2
λ_5^2	2
λ_1^2	12
T	-4

 $\beta_{\lambda_2}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$\lambda_2 T_\nu$	4
g_2^4	$\frac{9}{4}$
$g_1^2(\frac{3g_2^2}{2} - 3\lambda_2)$	1
$g_2^2\lambda_2$	-9
g_1^4	$\frac{3}{4}$
λ_3^2	4
$\lambda_3\lambda_4$	4
λ_4^2	2
λ_5^2	2
λ_2^2	12
T_ν	-4

 $\beta_{\lambda_3}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$\lambda_3(2T_\nu + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_e(Y_e)^\dagger) + 6 \text{Tr}(Y_u(Y_u)^\dagger) + 4\lambda_4)$	1
$\lambda_4(T_\nu + 3 \text{Tr}(Y_d(Y_d)^\dagger) + \text{Tr}(Y_e(Y_e)^\dagger) + 3 \text{Tr}(Y_u(Y_u)^\dagger))$	1
$\text{Tr}(h(h)^\dagger Y_e(Y_e)^\dagger)$	-2
g_2^4	$\frac{9}{4}$
$g_2^2(-9\lambda_3 - \frac{9\lambda_4}{2})$	1
g_1^4	$\frac{3}{4}$
$g_1^2(-3\lambda_3 - \frac{3\lambda_4}{2})$	1
λ_3^2	4
$\lambda_1(6\lambda_3 + 3\lambda_4)$	1
$\lambda_2(6\lambda_3 + 3\lambda_4)$	1
λ_4^2	4
λ_5^2	6

 $\beta_{\lambda_4}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$\lambda_3(T_\nu + 3 \text{Tr}(Y_d(Y_d)^\dagger) + \text{Tr}(Y_e(Y_e)^\dagger) + 3 \text{Tr}(Y_u(Y_u)^\dagger) + 8\lambda_4)$	1
$\lambda_4(2T_\nu + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_e(Y_e)^\dagger) + 6 \text{Tr}(Y_u(Y_u)^\dagger))$	1
$\text{Tr}(h(h)^\dagger Y_e(Y_e)^\dagger)$	2
g_2^4	$\frac{9}{8}$
$g_1^2(\frac{9g_2^2}{4} - \frac{3\lambda_3}{2} - 3\lambda_4)$	1
$g_2^2(-\frac{9\lambda_3}{2} - 9\lambda_4)$	1
g_1^4	$\frac{3}{8}$
λ_3^2	2
$\lambda_1(3\lambda_3 + 3\lambda_4)$	1
$\lambda_2(3\lambda_3 + 3\lambda_4)$	1
λ_4^2	5
λ_5^2	9

$\beta_{\lambda_5}^{(1)}$

$\mathcal{S}$	Scotogenic – dS2 – N ($\alpha = -1$)
$\lambda_5(2T_\nu + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_e(Y_e)^\dagger) + 6 \text{Tr}(Y_u(Y_u)^\dagger))$	1
$g_2^2\lambda_5$	-9
$g_1^2\lambda_5$	-3
$\lambda_3\lambda_5$	8
$\lambda_4\lambda_5$	12
$\lambda_5\lambda_1$	2
$\lambda_5\lambda_2$	2

Formal matching bridge

Matchete keeps the validated two-leg T3 topology, but those legs are two descriptions of the same physical scalar in this branch.